WORKED SOLUTION

STEP III 2023, Question 8

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The modulus makes the equation non-linear, but only mildly: on any stretch where y' keeps one sign, $|y'|$ is just y' or $-y'$ and the equation is an ordinary constant-coefficient one. So we follow the solution until y' hits zero, switch to the other linear equation, and match y and y' at the join. Part (iv) is the labour-saving trick: reflecting a solution, $x \mapsto c - x$, flips the sign of the first-derivative term, which is exactly the switch between the two linear equations.

Solution

Part (i)

With $\mathrm{f}(x) = x\mathrm{e}^{-x}$,

$$\mathrm{f}'(x) = (1-x)\mathrm{e}^{-x}, \qquad \mathrm{f}''(x) = -\mathrm{e}^{-x} - (1-x)\mathrm{e}^{-x} = (x-2)\mathrm{e}^{-x}.$$

Hence

$$\mathrm{f}'' + 2\mathrm{f}' + \mathrm{f} = [(x-2) + 2(1-x) + x]\mathrm{e}^{-x} = 0$$

for every real x. Also $\mathrm{f}(0) = 0$ and $\mathrm{f}'(0) = 1 \cdot \mathrm{e}^{0} = 1$, as required.

Since $\mathrm{e}^{-x} > 0$, the sign of $\mathrm{f}'(x)$ is the sign of $1 - x$, which is ≥ 0 for $x \leq 1$. So $\mathrm{f}'(x) \geq 0$ for $x \leq 1$.

Part (ii)

On $x \leq 1$ we have $\mathrm{f}' \geq 0$, so $|\mathrm{f}'| = \mathrm{f}'$ and the equation of (ii) is the same as the equation of (i). Thus f satisfies the new equation and its initial conditions on $x \leq 1$, and

$$\mathrm{g}_1(x) = x\mathrm{e}^{-x}.$$

At the join, $\mathrm{g}_1(1) = \mathrm{e}^{-1}$ and $\mathrm{g}_1'(1) = 0$. Continuous differentiability means g_2 must start with the same values: $\mathrm{g}_2(1) = \mathrm{e}^{-1}$, $\mathrm{g}_2'(1) = 0$.

Ask yourself Which sign does y' take just after $x = 1$?

At $x = 1$ the equation gives $y'' = -2|y'| - y = -\mathrm{e}^{-1} < 0$, so y' is decreasing through 0 and is negative just beyond $x = 1$. There $|y'| = -y'$, and the equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0.$$

The auxiliary equation $m^2 - 2m + 1 = 0$ has the repeated root $m = 1$, so $\mathrm{g}_2(x) = (A + Bx)\mathrm{e}^x$ and $\mathrm{g}_2'(x) = (A + B + Bx)\mathrm{e}^x$. The conditions at $x = 1$ give

$$(A + B)\mathrm{e} = \mathrm{e}^{-1}, \qquad (A + 2B)\mathrm{e} = 0,$$

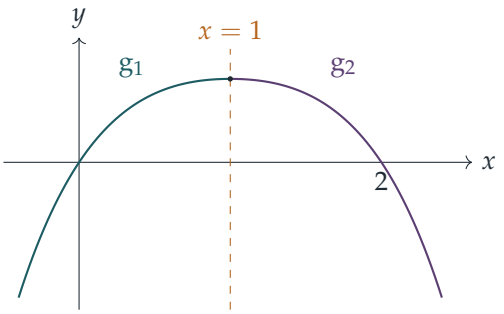
so $B = -\mathrm{e}^{-2}$ and $A = 2\mathrm{e}^{-2}$. Therefore

$$\mathrm{g}_2(x) = (2 - x)\mathrm{e}^{x-2}.$$

We must check this is consistent with the assumption $y' \leq 0$: $\mathrm{g}_2'(x) = (1 - x)\mathrm{e}^{x-2} \leq 0$ for all $x \geq 1$. So $|\mathrm{g}_2'| = -\mathrm{g}_2'$ throughout $x \geq 1$, and g_2 really does satisfy the modulus equation on the whole of $x \geq 1$.

Part (iii)

Replacing x by $2 - x$ in g_1 gives $(2 - x)\mathrm{e}^{-(2-x)} = (2 - x)\mathrm{e}^{x-2} = \mathrm{g}_2(x)$. So $\mathrm{g}_2(x) = \mathrm{g}_1(2 - x)$: the curve $y = \mathrm{g}_2(x)$ is the reflection of $y = \mathrm{g}_1(x)$ in the line $x = 1$.



Part (iv)

Let $Y(x) = \mathrm{k}(c - x)$ and write $u = c - x$. By the chain rule, with $\mathrm{d}u/\mathrm{d}x = -1$,

$$Y'(x) = -\mathrm{k}'(u), \qquad Y''(x) = \mathrm{k}''(u).$$

Hence

$$Y'' - pY' + qY = \mathrm{k}''(u) + p\,\mathrm{k}'(u) + q\,\mathrm{k}(u),$$

and the right-hand side is zero whenever $r \leq u \leq s$, because k solves the original equation there. Now $r \leq c - x \leq s$ is equivalent to

$$c - s \leq x \leq c - r,$$

so $y = \mathrm{k}(c - x)$ satisfies $y'' - py' + qy = 0$ on the interval $[c - s, c - r]$.

This explains (iii): with $p = 2$, $q = 1$, $\mathrm{k} = \mathrm{g}_1$ on $(-\infty, 1]$ and $c = 2$, the reflected function $\mathrm{g}_1(2 - x)$ solves $y'' - 2y' + y = 0$ on $[1, \infty)$.

Part (v)

With $\mathrm{h}(x) = \mathrm{e}^{-x} \sin x$,

$$\mathrm{h}'(x) = \mathrm{e}^{-x}(\cos x - \sin x) = \sqrt{2}\,\mathrm{e}^{-x} \cos\left(x + \tfrac{1}{4}\pi\right),$$

so $\mathrm{h}'(\frac{1}{4}\pi) = \sqrt{2}\,\mathrm{e}^{-\pi/4} \cos \frac{1}{2}\pi = 0$. (The second form also shows $\mathrm{h}' \geq 0$ on $[-\frac{3}{4}\pi, \frac{1}{4}\pi]$, as given, and that $\mathrm{h}'(-\frac{3}{4}\pi) = 0$, which we shall need.)

The plan mirrors part (ii). On an interval where $y' \leq 0$ the equation is $y'' - 2y' + 2y = 0$; where $y' \geq 0$ it is $y'' + 2y' + 2y = 0$, which h satisfies.

(a) At $x = \frac{1}{4}\pi$ we have $y = \mathrm{h}(\frac{1}{4}\pi) = \frac{1}{\sqrt{2}}\mathrm{e}^{-\pi/4} > 0$ and $y' = 0$, so $y'' = -2y < 0$ and y' turns negative. We therefore want a solution of $y'' - 2y' + 2y = 0$, and part (iv) with $p = q = 2$ hands us one: $y = \mathrm{h}(c - x)$ works for $c - \frac{1}{4}\pi \leq x \leq c + \frac{3}{4}\pi$. Choosing $c = \frac{1}{2}\pi$ makes this interval exactly $[\frac{1}{4}\pi, \frac{5}{4}\pi]$, and

$$y = \mathrm{h}\left(\tfrac{1}{2}\pi - x\right) = \mathrm{e}^{x - \pi/2} \sin\left(\tfrac{1}{2}\pi - x\right) = \mathrm{e}^{x - \pi/2} \cos x.$$

? **Ask yourself** How do we know this is the right solution, and not just a solution?

It matches at the join: $\mathrm{h}(\frac{1}{2}\pi - \frac{1}{4}\pi) = \mathrm{h}(\frac{1}{4}\pi)$, and its derivative $-\mathrm{h}'(\frac{1}{2}\pi - x)$ equals $-\mathrm{h}'(\frac{1}{4}\pi) = 0$ at $x = \frac{1}{4}\pi$. So y and y' are continuous there. It is also self-consistent: for $x \in [\frac{1}{4}\pi, \frac{5}{4}\pi]$ we have $\frac{1}{2}\pi - x \in [-\frac{3}{4}\pi, \frac{1}{4}\pi]$, where $\mathrm{h}' \geq 0$, so $y' = -\mathrm{h}'(\frac{1}{2}\pi - x) \leq 0$ and $|y'| = -y'$ throughout. Since a linear equation with given y and y' at a point has only one solution, this is the continuation. So on $\frac{1}{4}\pi \leq x \leq \frac{5}{4}\pi$,

$$y = \mathrm{e}^{x - \pi/2} \cos x.$$

(b) At $x = \frac{5}{4}\pi$: $y = \mathrm{e}^{3\pi/4} \cos \frac{5}{4}\pi = -\frac{1}{\sqrt{2}}\mathrm{e}^{3\pi/4} < 0$, and $y' = -\mathrm{h}'(-\frac{3}{4}\pi) = 0$. Now $y'' = -2y > 0$, so y' turns positive and the equation reverts to $y'' + 2y' + 2y = 0$.

This equation has constant coefficients, so a translate of a solution is a solution. Since $\sin(x - 2\pi) = \sin x$, the function $\mathrm{h}(x - 2\pi) = \mathrm{e}^{2\pi - x} \sin x$ solves it, and on $[\frac{5}{4}\pi, \frac{9}{4}\pi]$ its argument $x - 2\pi$ runs over $[-\frac{3}{4}\pi, \frac{1}{4}\pi]$, where $\mathrm{h}' \geq 0$, so it is consistent with $y' \geq 0$. Check the join at $x = \frac{5}{4}\pi$:

$$\mathrm{e}^{2\pi - 5\pi/4} \sin \tfrac{5}{4}\pi = -\tfrac{1}{\sqrt{2}}\mathrm{e}^{3\pi/4}, \qquad \mathrm{h}'(\tfrac{5}{4}\pi - 2\pi) = \mathrm{h}'(-\tfrac{3}{4}\pi) = 0,$$

both matching. So on $\frac{5}{4}\pi \leq x \leq \frac{9}{4}\pi$,

$$y = \mathrm{e}^{2\pi - x} \sin x.$$

At $x = \frac{9}{4}\pi$ this is $e^{-\pi/4}\sin\frac{1}{4}\pi = h(\frac{1}{4}\pi)$ with zero derivative: the solution is back where it was at $\frac{1}{4}\pi$, so for $x \geq \frac{1}{4}\pi$ it is periodic with period 2π.

Checks

- The pieces join smoothly by construction, but it is worth a direct look: at $x = \frac{5}{4}\pi$ both (a) and (b) give $y = -\frac{1}{\sqrt{2}}e^{3\pi/4} \approx -7.4605$, with zero slope.
- Stepping the modulus equations forward numerically from $y(0) = 0$, $y'(0) = 1$ reproduces g_1, g_2 on $[0, 4]$ and the three pieces of (v) on $[0, \frac{9}{4}\pi]$ to within 10^{-9}.
- Differentiate (a) directly: $y' = e^{x-\pi/2}(\cos x - \sin x)$, $y'' = -2e^{x-\pi/2}\sin x$, and $y'' - 2y' + 2y = 0$ as required.

Common slips

- Keeping $|y'| = y'$ after the turning point: f itself is not the solution beyond $x = 1$.
- Forgetting to check that the new piece keeps y' of the assumed sign over the whole interval.
- In (iv), giving the interval as $[r, s]$; the reflection maps it to $[c - s, c - r]$.

Beyond STEP

Equations like these, whose right-hand side is smooth except across a switching surface (here $y' = 0$), are called piecewise-smooth or switching systems; the solution is built by gluing linear pieces, and existence and uniqueness still follow from the Picard–Lindelöf theorem because $|y'|$ is Lipschitz. The 2π-periodicity in (v) is something a linear damped equation could never produce: with $E = \frac{1}{2}y'^2 + y^2$ we get $dE/dx = -2y'|y'|$, so the "damping" term removes energy while y rises and puts it back while y falls. Boyce and DiPrima, *Elementary Differential Equations*, covers the linear theory and uniqueness.

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